Learning airline route preferences from flight-plan revisions

Ramon Dalmau[a,*], David Perez[b], Franck Ballerini[a], Seddik Belkoura[b], Vincent Taverniers[b], Gratiel Marius Marin[b], Robin Deransy[a], Benjamin Cramet[c], Grigori Gustin[c]

[a]*EUROCONTROL Innovation Hub, Brétigny-sur-Orge, France*
[b]*EUROCONTROL Maastricht Upper Area Control Centre, Maastricht, the Netherlands*
[c]*EUROCONTROL Headquarters, Brussels, Belgium*

Abstract

Every flight-plan revision records a choice an airline actually made: the route it dropped and the route it filed instead. The aim is to cut carbon dioxide by offering cheaper routes only where they would be taken. Eighteen months of European traffic contain 1.48 million such pairs. A pairwise ranking model fitted on 1 134 000 of them identifies the chosen route in 68.1 % of held-out decisions. Planning cost points the wrong way: preferring the route with less planned fuel scores 45.0 %, below chance, so the strongest simple rule is its inverse, at 55.0 %. A channel must tell the flights that would take the cheaper route from those that would not, and the model does: it is right 94.9 % of the time on the fifth of cases it is surest about. Aimed at 80 % precision and measured on a held-out quarter, such a channel reached 79.2 %, its proposals carrying a planned-fuel saving of 20.1 kilotonnes a quarter on routes the airlines went on to file, or 222 to 254 kilotonnes of CO_2 a year on two annualisation bases. Refitted on earlier data and watched for six months, the model ages by proposing less, not erring more. The clearest gap is the flights that never revise, though a much scarcer set of pairs, each matching a kept route against one a similar flight flew, corrects nearly all of the model's bias towards change. Even so, airline route acceptance is learnable from operational records alone.

*Corresponding author.
Email address: `ramon.dalmau-codina@eurocontrol.int` (Ramon Dalmau)

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1. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the airline subsequently re-filed. The new route shed that delay completely, and it cost the airline on each of the three counts an airline pays for directly: 39 minutes of flight time, 1 397 kg of extra planned fuel, and 466 euros more in en-route charges. Both routes are held in the network's records and so, implicitly, is the airline's judgement that the delay would have cost it more than the detour did.

That judgement is the quantity this paper models. A flight plan is not filed once and left alone: an airline may revise it at any time until the aircraft leaves, and the response of the network is only one of the reasons for doing so. Attributed delay of the kind the opening flight was carrying is the most visible prompt. An aircraft arriving late from its previous leg is another, since the delay it brings with it consumes the slack the rotation was holding; and an updated forecast is a third, since changed winds change the time and the fuel that every candidate route would need. A route may also simply cease to be filable, in which case the revision is not a choice at all. What the archive records is what changed rather than why, and it does not separate these cases. Every revision nonetheless leaves the same trace, namely the route replaced and the route that replaced it, and therefore a preference that the airline expressed with its own filing system rather than in answer to a survey.

The scale of what is at stake is published every year. Air traffic in Europe is planned across national borders by a single network function, and an independent commission reviews that function's performance annually, reporting how much further the traffic flies than a direct reference between the points at which it enters and leaves the airspace (Performance Review Commission, 2026). Not all of that distance is avoidable: airspace that cannot take all the traffic that wants it has to be protected, and a flight routed around it flies further, so part of the figure is what respecting a limited capacity costs. What this paper takes up is whether any of the rest is a cheaper route that the airline would have been willing to fly and did not file. That such routes exist is assumed here rather than shown, and Section 9 returns to the assumption; a route closed by a restriction or an unavailable conditional route is not one of them, since it could never have been filed. Candidate routes are not hard to come by. What is hard is knowing which candidate this particular airline would accept, on this day, with this rotation to protect, and knowing it while there is still time to file.

What the network lacks is therefore a proposal channel, that is, a service that would offer an airline an alternative route that burns less fuel and that the model judges the airline would accept, inside the planning window and before the flight departs. The timing is not free: two of the model's strongest inputs, the attributed delay and the airspace constraints behind it, exist only once a plan has been filed, so the channel would speak after a first filing rather than before one, and Section 9 returns to what that costs.

The objective of such a channel, and of this paper, is twofold, and the first half of it is environmental: a route that burns less planned fuel emits less carbon dioxide when it is flown. The second half is that the saving must not be bought at the airline's expense, because a route that saves a tonne of fuel while costing an hour of delay or a broken rotation is not a saving any airline will take. What this paper supplies is the instrument that allows the two to be pursued together, namely a way of ranking candidate routes by how likely the airline is to accept them, so that the environmentally cheaper proposals can be aimed at the flights on which they are also cheap in the airline's own terms. Needless to say, proposing routes that airlines reject is worse than proposing nothing at all, because a channel that is ignored is a channel that is eventually switched off.

A preliminary version of this work was presented at the SESAR Innovation Days 2026, where the question was whether route acceptance can be predicted at all. The present paper is the treatment of the channel: what it would address, at which operating points, for which airlines, and how the figure survives the design decisions behind it. It adds the baseline ladder and the calibration table behind those results, the channel simulation across four operating points, the concentration of its value across operators and its sensitivity to the bias correction, and a retraining-horizon experiment in which the model is trained on one year and watched month by month over the following half year.

The contribution of this paper is fourfold. First, it shows that airline route acceptance is learnable at network scale from revision events alone, and that the confidence of the resulting model is calibrated well enough to support a chosen operating point rather than a single accuracy figure. Secondly, it establishes that the learned preference actively contradicts planning cost: every rule of the form "prefer the cheaper route" scores below chance on held-out revisions, which is the cleanest evidence available that filing behaviour carries structure a cost model cannot represent. Thirdly, it quantifies what a proposal channel built upon such a model could address, using the airlines' own subsequent filings as ground truth. That quantification also reports how the value concentrates across airlines and how sensitive it is to the design decisions. Fourthly, it measures how quickly the model ages, which is the question a deployment plan needs answered and the one a single train–test split cannot address.

The remainder of this manuscript is organised as follows. Section 2 explains why a pair of routes drawn from a revision constitutes a preference observation, names the two operational use cases that such an observation is meant to serve, and sets out why revision pairs carry a stronger label than pairs assembled from generated candidates. Section 3 positions the work within the literature on rerouting, route choice and learning from implicit feedback. Section 4 describes the archive, the construction of the training pairs and the encoding that keeps leakage out of them, and Section 5 the model, its training and its calibration. Section 6 reports predictive performance and its structure, Section 7 the proposal-channel simulation, and Section 8 the retraining-horizon experiment. Section 9 collects the threats to validity, and Section 10 concludes.

2. Background

This section describes fundamental concepts that are not a contribution of this work, but that are needed to follow the sections that come after it. Thus, a reader who is already familiar with the European filing process and with discrete choice may move on to Section 3.

2.1. Flight-plan revisions as preference data

In the European network a flight plan is filed before departure and may be revised at any time until the flight leaves. Every filing is a message, and so is every later change, and the network keeps them all, so that what it holds for a flight is the whole sequence of messages sent for it. A route change is accordingly a pair of consecutive messages rather than a record of its own: the message that changed the horizontal route of the plan, and the message that preceded it. Those two messages together describe a decision, because the airline had committed to the earlier route and then filed the later one for the same flight on the same day, which makes the later route the one it chose and the earlier one the route it abandoned. That the pair is assembled from two messages sent at different times is not a detail of bookkeeping: it shapes the encoding, and Section 4.3 returns to it.

The sharpest of those decisions, and the ones in which the terms of the trade are clearest, are made under flow management, and the term this paper uses for the measure itself needs stating plainly. When the traffic expected in a volume of airspace exceeds what the controllers there can safely handle, whether because of demand, staffing, weather or equipment, a *regulation* is placed on that volume for a period, and the flights planning to cross it are given calculated take-off times at the rate the volume can absorb. A flight held back in this way is given a take-off time later than the one it requested, and the difference between the two is its *attributed ATFM delay*: the aircraft waits on the ground rather than in the air. Not every flight crossing a regulated volume is delayed, since a lightly binding regulation absorbs most of its traffic at the requested times, and a flight caught by several regulations is counted against the one that penalises it most.

An airline facing such a delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel, en-route charges and the departure slot it already holds, and neither answer is universally correct. For instance, a flight with a tight aircraft rotation (i.e., a schedule in which the same aircraft flies several legs in the day, so that delay on one leg propagates to the next) may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. This is why the decision is worth learning rather than deriving: the trade depends on airline-specific and flight-specific costs that the network does not observe. Published European reference values (Cook and Tanner, 2015) give the order of magnitude of the cost of delay and are widely used for analysis, but they are not the cost function any particular airline applies, and what an individual operator would charge itself for forty minutes is not known outside it.

Most revisions, though, are not a response to flow management at all: in seven pairs in ten neither route carries a regulation, and Section 4 reports what that majority looks like. A dispatcher re-planning after a forecast update, or after the aircraft's earlier legs have moved, leaves the same kind of trace as one escaping a delay, and the archive records what changed rather than why.

2.2. Two operational use cases

The two uses the model is intended for share exactly one quantity, namely the probability that a particular airline would accept a particular alternative for a particular flight, and differ in almost everything else.

The first use case is a proposal channel: for a flight whose filed route has an alternative that burns less planned fuel, the channel proposes that alternative to the airline, but only when the model judges that the airline would accept it. What is on offer is not a route the airline could not have found, but rather a route it did not evaluate in time: flight planning happens against a deadline, the candidate routes are many, and an alternative that is both cheaper and acceptable can pass unnoticed simply because nobody looked at it before the plan was filed. This is the use case that Section 7 simulates, with the airlines' own later filings standing in for the answers a real channel would collect.

The second use case is the release of an overloaded volume of airspace, meaning the removal of enough flights from it to bring the demand back within the capacity declared for it. Such an overload is what prompts a regulation in the first place, and the flights it catches may reroute of their own accord. A channel of the kind just described could instead put a specific alternative to specific flights, and would then have to decide which flights to approach first. The formulations that make such a choice optimise a network objective and do not model whether the airline would fly what it is offered, while the collaborative variants treat the airline's preference as an input the airline must supply (Section 3). A calibrated acceptance probability supplies that input for every flight, whether or not its airline has declared anything, and it changes the ordering: approaching first those flights for which an acceptable and environmentally cheaper alternative exists means that one action relieves the overload, reduces the fuel burnt on the flight, and asks the least of the airline. Acceptance remains voluntary, and this is a reordering of a decision taken in any case rather than a new intervention.

Neither use case has been tested end to end. The first is simulated, in the sense that every proposal is checked against what the airline subsequently filed, so the figures reported herein are what a channel would have achieved had the airlines behaved as they in fact behaved; no airline was ever asked anything. The second is enabled rather than demonstrated, in that the model supplies the input the network-side formulations lack while no overload-release experiment is run here.

Both uses would, once deployed, produce the labels they need, since every proposal answered is a new observation of exactly the kind the model is trained on. Those answers would be a biased sample even so, because an airline can only answer the proposals it is sent and those are the proposals about which the model was already confident. Measuring that bias is as much the purpose of a trial in which real proposals are made and the airlines' answers logged (Section 10) as measuring acceptance is.

2.3. Revealed preference and the construction of pairs

Readers of this journal will recognise the setting as a route-choice problem, a family with a long history in travel-demand modelling (McFadden, 1974; Ben-Akiva and Lerman, 1985; Train, 2009). The observation unit here is the choice between exactly two alternatives that the decision maker itself constructed, which sidesteps the choice-set generation problem that route-choice models must otherwise solve, and the estimator is a gradient-boosted ranker rather than a parametric logit. Roughly speaking, the model of this paper is a conditional logit whose utility function is learned nonparametrically. The training objective penalises the model according to how far the abandoned route's score sits above the chosen one's, and keeps pressing even once the order is right, exactly as a binary logit on utility differences would; what differs is that the score is an arbitrary function of the features rather than a linear index. What is given up is the economic interpretability of coefficients; what is gained is the ability to absorb the waypoint sequence itself, which Section 6.5 shows to be the largest single term in what moves the model, though smaller than the four cost indicators taken together. The economic tradition supplies the framing that matters here, namely revealed rather than stated preference (Samuelson, 1938): every label in this paper is something an airline did, not something it said it would do.

Revealed preference has a direct consequence for how a training pair ought to be built, and it is the decision on which everything that follows rests. The natural way to build such a pair without revisions is to take the filed plan as preferred and a set of generated routes as less preferred. The authors' own earlier work on ranking flight routes, whose purpose was to predict the flight plans not yet filed so that traffic demand could be estimated earlier than the filing horizon allows, was constructed in that way (Dalmau et al., 2024). Its training pairs take the filed plan as preferred and a set of generated proposals (up to eleven route options produced by the Network Manager, the network-level flow-management function of EUROCONTROL) as less preferred, and on the pre-tactical set none of the latter need match the filed route at all. The weakness of such a label is easiest to see in a single case. Suppose an airline filed route A, and that the training pair ranks A above a route B that the analyst generated. Filing A shows only that A was acceptable. Nothing in the record shows that B ever reached the dispatcher who planned that flight, still less that it was compared with A and rejected, so the pair asserts a comparison that may never have taken place. A model can satisfy such a label by learning what a filed plan looks like rather than what an airline wants. For instance, a generated candidate may join published segments in a way no dispatcher would ever file, so that telling it from a filed plan can be done on the shape of the route alone, without knowing anything about the airline that would fly it. Recognising filed plans and ranking routes by preference are different skills, and an accuracy figure cannot distinguish them.

A revision pair carries no such doubt, and Fig. 1 draws the two constructions side by side on one city pair. Both routes of a revision reached the dispatcher's screen, since the airline filed each of them in turn, for the same flight, on the same day, so the route labelled less preferred is not an alternative constructed after the fact but one the airline had committed to and then dropped. Whatever else was happening, both routes were under the airline's own consideration, and it moved from one to the other. No dataset offers both label sources for the same flights, so this is an argument about how the labels are built rather than a measured comparison.

The question also narrows, in that the model is not asked what an airline will file but which of two routes it prefers, which is a strictly easier question and precisely the one a proposal channel needs answered. It should be noted, however, that the pair is the unit of *training* and not a limit on use. The trained model gives a score to one route at a time, against the other candidates on offer for the same flight: the encoding of Section 4.3 expresses each indicator as the amount by which that route exceeds the cheapest candidate, so what the model needs is the candidate set, not a pair. A flight offered five routes is therefore scored five times against the cheapest of the five, and a flight offered fifty is scored fifty times, with the pairwise objective doing nothing more than teaching the model where to place those scores relative to one another. Every figure in this paper is nonetheless measured on pairs, because pairs are what the archive records.

(a) candidates (b) a revision

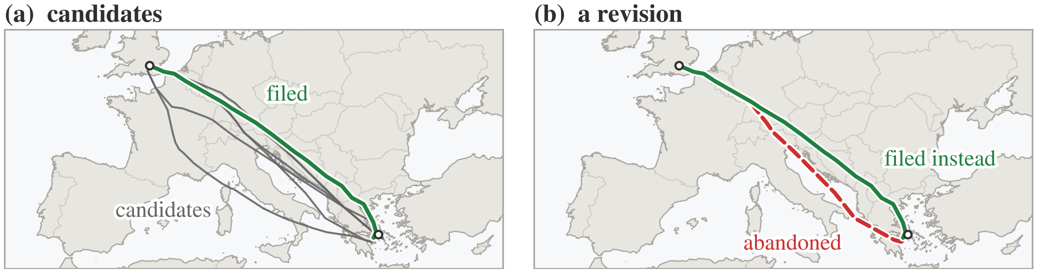

Figure 1: Two ways to build a training pair, on one city pair; each route is labelled by its role where it is drawn. Panel (a): the filed route (solid green) is ranked above candidates from a route generator (grey), which the airline may never have evaluated; those drawn here are other routes observed on the same city pair. Panel (b): a revision supplies two routes that the same airline filed for the same flight hours apart, one of which it abandoned. Only the right-hand pair records a preference the airline actually expressed.

The price to be paid is that revisions are not a random sample of flights, but rather flights that changed, which introduces a bias of its own. Rather than assuming that bias away, Section 6.3 measures it.

3. Related work

Three threads of literature meet in this paper, and the first of them, rerouting in air traffic flow management, has largely approached the problem from the network side. Over two decades ago, rerouting was formulated as a dynamic network flow problem (Bertsimas and Patterson, 2000), and a stochastic extension later rerouted flights as capacity evolved (Mukherjee and Hansen, 2009). Both optimise an objective defined by the network and assume, in effect, that the resulting assignment is executed, so that the willingness of the airline to fly it is not modelled. The value of airline collaboration in flow management has been examined (Castelli et al., 2012), which motivates the proposal channel simulated herein but does not address which proposals would be taken up. The closest work in this thread incorporates airline preferences into collaborative rerouting decisions and shows that accounting for them changes which reroutes are selected (Murça, 2018; Xu et al., 2020), as does the collaborative trajectory options programme, in which the airline attaches to each alternative route it submits a cost relative to the option it prefers (Kulkarni, 2018). In all of these the preference is an input the airline must supply, whereas this paper learns the preference function itself, at network scale, from what airlines did rather than from what they stated, and so supplies the input for every flight whether or not its airline has declared anything. Preferences have on occasion been estimated rather than collected, since trajectory options and the preferences over them have been derived from historical traffic by clustering and classification and then passed to a demand-capacity optimisation (De Giovanni et al., 2024), while the probability that a regulated flight reroutes of its own accord to escape delay has been predicted so that a flow manager may weigh the consequences of a regulation before activating it (Dalmau, 2022). Both estimate what airlines tend to do in the absence of a proposal, whereas releasing an overloaded volume requires knowing whether a particular flight would accept a particular alternative, which is the quantity the present model supplies. Beyond the research literature, airlines already express flight-level priorities within a regulation through the user-driven prioritisation process (Pilon et al., 2021; Ruiz et al., 2019), an operational mechanism concerned with the ordering of delay among an airline's own flights rather than with route choice; a preference model of this kind addresses a different quantity and would complement it rather than replace it.

The second thread is economic. Earlier than any of the work above, route choice in Europe was studied as a trade between fuel and en-route charges (Delgado, 2015), which establishes that the choice is economic and therefore learnable, but which uses no label in which the airline rejected an alternative it had itself filed. The discrete-choice tradition discussed in Section 2.3 supplies the theoretical frame for such labels; what it has lacked in this domain is a data source in which both alternatives were genuinely available to the decision maker. Revisions are exactly that source.

The third thread is methodological. Learning a ranking function from pairs is itself standard (Burges et al., 2005; Liu, 2009), and learning preferences from logged behaviour rather than from elicited judgements was established in information retrieval, where clicks on search results were shown to yield reliable relative (though not absolute) relevance labels (Joachims, 2002). The revision archive is the aviation analogue of a click log: nobody was asked anything, and each record is trustworthy only as a comparison between the alternatives that were actually available. The same literature also supplies the caution that this paper inherits, namely that logged data reflects the behaviour of the system that logged it, a point returned to in Section 9.

The direct predecessor of this work (Dalmau et al., 2024) established the framing adopted here, namely a gradient-boosted ranker that scores routes by their indicators in context, and reported the difficulty that motivates the present paper, namely that its training pairs rank the filed plan above candidates the airline may never have examined. Section 2.3 sets out that argument and the label construction that follows from it.

4. Data and pair construction

Figure 2 shows the procedure end to end, from the archive to a proposal. This section covers its first three stages, namely the archive, the pairs built from it and the encoding under which they reach the model; the model and the calibration follow in Section 5.

4.1. The archive

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs. Each route is described by twenty-three features, summarised in Table 1, one of which is in-house shorthand and worth spelling out: the protected volume is the volume of airspace whose capacity the regulation was declared to protect.

"Airline" is used loosely throughout for any aircraft operator, since the archive contains business, cargo and non-scheduled traffic as well as scheduled airlines, and the figures say "operator" where the count is over all of them. All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements. Operator identity enters the model as a categorical feature only, and no operator or airspace volume is named anywhere in this paper. A small number of individual flights are reported as worked examples, and nothing beyond what each example needs is given about them.

What those 1.48 million decisions look like matters before any model touches them, and Table 2 breaks the archive down by the regulation context in which each revision was made.

Seven revisions in ten carry no regulation on either route, so most of the archive records ordinary flight-plan housekeeping rather than escapes from regulations, and the median fuel difference within those pairs is a single kilogramme. Where a revision does escape a regulation it pays for the escape, since only 28.1 % of escape revisions move to a route that burns less fuel and the median move costs 101 kg and 2.1 minutes. The mirror image of that

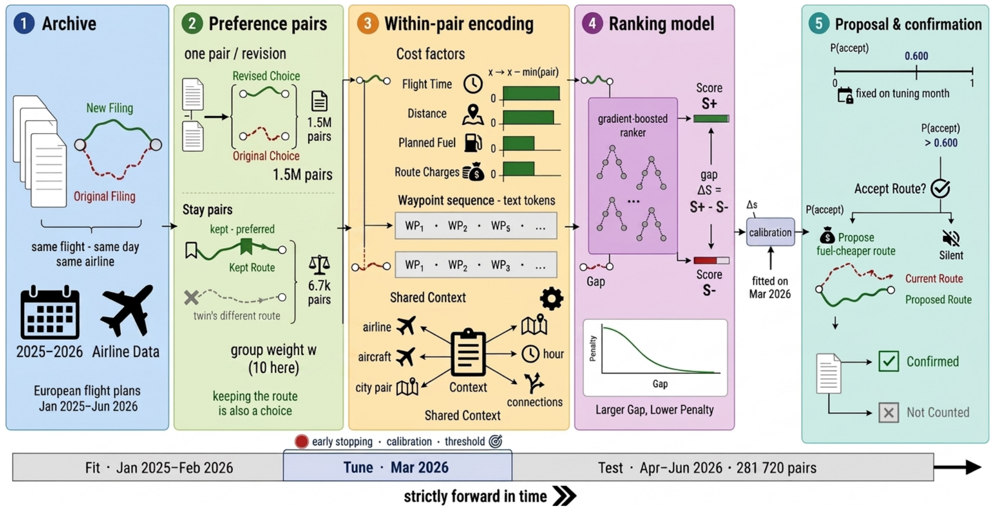

Figure 2: How a filing archive becomes a proposal. Each revision yields two routes for the same flight, exactly one of which the airline chose, giving 1.48 million revision pairs; stay pairs (flights that kept their route, Section 4.2) invert that construction and contribute 6 672 pairs in all, of which the 5 299 that fall in the fitting period enter at a group weight of 10 so that the smaller set is not swamped by the larger one, 280 fall in the tuning month and 1 093 in the test months; the value is chosen by the sweep of Section 6.3. Encoding takes differences within the pair rather than absolute values. The ranking model returns one score per route, drawn as the two bars in that box, S^+ for the higher and S^- for the lower; the inset below them is the training objective, whose penalty falls as the gap between the two widens and never reaches zero. That gap becomes an acceptance probability, compared with a threshold fixed in advance at 0.600, which is the only condition on a proposal. The bar along the foot of the figure is the temporal split of the archive: the model is fitted on the earliest stretch, tuned on the next month and tested on the latest, so no pair is ever scored by a model that saw it. The bars of the third stage are one illustrative pair, in which the route the airline chose is the costlier on each indicator; that is the commoner direction on this archive (Section 6.1) rather than the invariable one.

escape is the smallest context of all, the 1.4 % of revisions that took on delay, which is the only context in this eighteen-month table in which the share of fuel-saving moves exceeds one half, because a flight that accepts delay tends to be paid in fuel for it. Everywhere else the share stays below one half, which already suggests that fuel is not what filing behaviour optimises, an observation that Section 6.1 sharpens into a measured result. Figure 3 shows two revisions drawn from the archive, an escape and a fuel saver.

Table 1: What the model sees for each route; some rows summarise several features (twenty-three in all). The first two groups and the last belong to the route and differ within a pair, the indicators entering as differences from the pair minimum; the rotation and context rows belong to the flight and carry the same value on both routes, the one exception being the outbound connection time, which is shortened on the slower route by the difference in flight time and is therefore route-dependent after all.

group	content
indicators	flight time, distance, planned fuel, en-route charges (the fees an airspace user pays for crossing each charging zone)
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume
rotation	inbound and outbound connection times and states
context	airline, aircraft type, city pair, hour, weekday, month, time to departure
route	the ordered waypoint sequence, as text

Table 2: The revision archive by decision context, over all eighteen months. A pair is counted as regulated here when a regulation was attributed to the flight while it was filed on that route, which leaves the delay itself free to be zero; Table 5 sorts the held-out quarter by the delay instead, and the two conventions do not give the same counts. Unregulated pairs carry no regulation on either route, escapes carry one on the abandoned route only, the next two contexts carry one on both, and in the final context only the newly filed route is regulated. The median fuel difference is between the two routes of a pair; the final column is the share of pairs in which the newly filed route burns less planned fuel than the one it replaced.

context	pairs	share [%]	median Δfuel (new − old) [kg]	new route saves fuel [%]
unregulated	1 047 228	70.6	1	46.2
escaped	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
took on delay	21 337	1.4	−5	51.8

4.2. Stay pairs

Every revision pair is, by construction, a flight that *did* change route, and that creates a trap. A model trained only on such events learns that change is

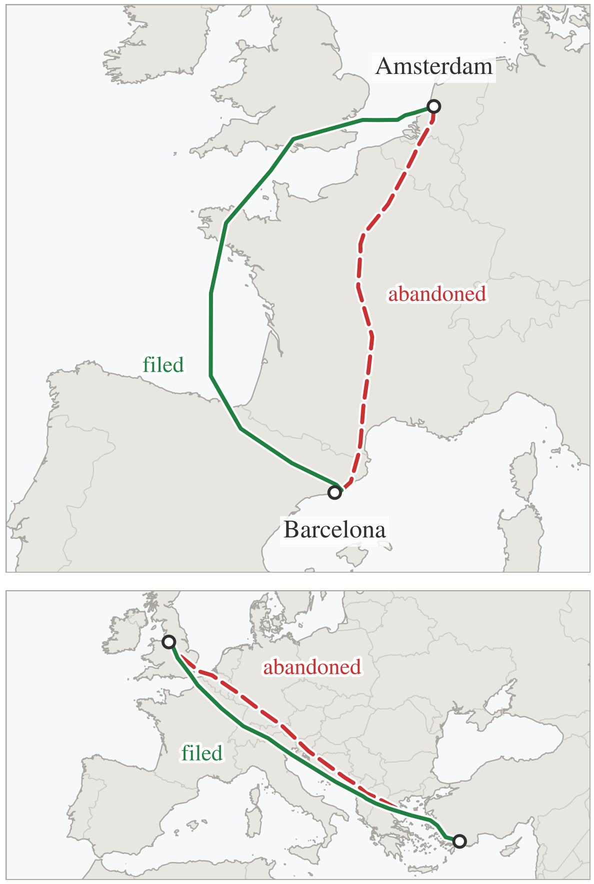

Figure 3: Two revisions from the archive, in the visual language used throughout this paper: the abandoned route is dashed red and its replacement solid green, each labelled on the track itself. Top: an escape, in which the airline paid fuel and time to leave a regulated volume; the volume itself is not drawn, since naming the airspace would identify it, and the two airports are named because this city pair is the running example of Section 1. Bottom: a fuel saver, in which the replacement was cheaper outright; its endpoints are left unnamed, that running example being the only city pair this paper identifies. The first kind accounts for a quarter of the regulated archive; the second is what a proposal channel exists to surface earlier.

what happens, and recommends it far too eagerly against traffic that mostly stays, by a margin that Section 6.3 measures. Put another way, the archive answers one question only, namely: when an airline changed its route, which route did it move to? It cannot answer the complementary question, because a flight that kept its route generates no record of the alternative that it declined.

A second, much smaller dataset was therefore constructed, in which *keeping* the route is the revealed choice. Let us consider two flights on the same day, with the same origin, destination and aircraft type, that flew different routes and neither of which revised its plan, and let us suppose that one of them was caught by a regulation while the other was not. The regulated flight had a reason to move and an alternative that was demonstrably flyable that day, namely the route its twin was flying, and yet it stayed; its route is accordingly labelled as preferred, and that of its twin as the alternative it declined. What the twin does not establish is that the same routing was open to the stayed flight at its own departure hour, since route availability in Europe is conditioned on the time of day, and that limit sits alongside the regulation confound described next. The conditions are restrictive and the yield is correspondingly low: 6 672 such *stay pairs* (pairs in which the preferred route is the one that was kept rather than the one that replaced it), of which 1 093 fall in the test months, against 1.48 million revision pairs. The two are used together, with stay pairs entering the fitting set at an increased group weight (the pair's weight in the ranking loss).

The stay pairs also carry evidence of their own, and it concerns the price airlines put on staying where they are. The flight that stayed faced a median peak delay (the largest delay attributed to it at any point before departure) of 31 minutes, and in 47.5 % of the pairs the route it kept burns *more* planned fuel than the alternative its twin was flying (the median difference is 3 kg in the kept route's favour, i.e. essentially nil). Airlines therefore accept delay even when a demonstrated, comparably priced alternative exists, which is revealed preference for stability itself, and precisely the behaviour an over-eager proposal channel would violate.

One property of this construction has to be stated alongside it, because it bounds what the stay pairs can ever be used to measure. The regulated flight is always the one whose route was kept, so regulation status identifies the labelled answer. Concretely, in all 1 093 stay pairs of the test months the kept route carries a positive attributed delay and at least one regulation, and the alternative carries neither, with no exceptions. A rule as crude as "prefer the route of the flight that was regulated" is therefore right on every stay pair without examining a route at all. Whether the model takes that shortcut is a question the archive can answer, because it also contains the mirror population in which the same rule is always wrong; Section 6.3 puts the two side by side.

4.3. Encoding and leakage control

Treating a revision as a labelled preference pair requires care, because the two routes are separated in time as well as in geography: the replacement is by construction the later message, and any feature that drifts with time can therefore identify the answer without saying anything at all about the quality of the route. This subsection is devoted to that problem, because on this archive it is not a technicality: a model built without attention to it would report excellent figures while knowing nothing whatever about routes.

One pair from the held-out quarter shows the size of the problem. Its original plan was filed at 00:30 and the plan that replaced it at 10:56, for a flight due off the blocks at about 12:30. Left as filed, the time remaining to departure would read 720 minutes on the abandoned route against 94 on the one that replaced it. That gap is not an outlier in kind: across the held-out quarter the median distance between the two filings is 33 minutes, a quarter of the pairs are more than an hour and a half apart, and the revision is the later message in all but a thousandth of them, where the two filings carry the same recorded time. A model given the feature in its raw form could therefore answer "the one filed later" and be right on essentially every pair while knowing nothing whatever about routes. This is leakage rather than learning, and it would be invisible in the accuracy figure: the model would score close to perfectly on held-out revisions, for the empty reason that held-out revisions carry the same timestamp asymmetry as the training ones. To remove the shortcut, all flight-level attributes were harmonised across the pair so that only route-level quantities differ, and the filing-time feature was collapsed to the pair minimum, which gives it an identical value on both routes. The stay pairs of Section 4.2 pass through the same step, the values of the flight that stayed being copied to the other row, so that in both populations only route-level quantities separate the two candidates.

A subtler shortcut hides in the absolute indicator values. Let us consider a 400 km flight on which the two routes burn 2 000 and 2 300 kg. A model shown the raw magnitudes learns, inevitably, what 2 000 kg means on this kind of sector, and with it everything that correlates with sector length: which airlines fly it, at which hours, with which aircraft. Some of that is legitimate context, but it also hands the model a way to answer from the flight rather than from the comparison, and it generalises poorly to sectors of other lengths. The within-pair encoding subtracts the pair minimum from each indicator, so that what reaches the model is 0 against 300 kg, a pure comparison that carries the same meaning whether the flight is 400 or 4 000 km.

4.4. Splits

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026 (1 134 064 pairs), tuned on March 2026 (67 349), and evaluated once on April to June 2026 (281 720), the held-out quarter cited throughout; the three account for all 1.48 million pairs of the archive. Every threshold, calibrator and stopping decision is taken on the tuning month alone, flights appearing in the tuning or evaluation months are removed from the fitting set, and eleven automated checks verify split disjointness, feature construction and the pair invariants. A second split, used only in Section 8, ends earlier so that the model can be watched for six months beyond its training cutoff: it is fitted on January to October 2025, tuned on November and December 2025, and evaluated month by month on January to June 2026. It is a shorter window whose tuning cutoff falls three months sooner, not the same window moved back a year.

5. Model, training and calibration

This section covers the remaining stages of Fig. 2: the scoring model, the objective it is trained under, and the calibration step that turns its scores into acceptance probabilities. The design follows the predecessor system (Dalmau et al., 2024) except where stated, and the emphasis here is on the two decisions that the results show to matter, namely the within-pair encoding and the stay-pair mixture.

The operational question is comparative: given two routes for one flight, which of them does the airline prefer? Accordingly, the model gives each route a single number, which is not a probability and has no units, and which is only meaningful next to the number given to another route for the same flight. Training pushes those numbers apart in the right direction. For every historical pair the model pays a penalty, largest when the abandoned route outscores the chosen one; the penalty falls as the chosen route's lead grows and never quite reaches zero, so the model keeps pressing even once the order is right (Burges et al., 2005). Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit, and which is also what connects the estimator to the conditional-logit tradition of Section 2.3.

Gradient-boosted decision trees (Prokhorenkova et al., 2017) are used as the scoring function, at a depth of 8, with the learning rate selected automatically and with early stopping on the tuning month. The waypoint sequence is carried as text and encoded as a bag of tokens, which is to say that each waypoint identifier is treated as a word and the route as the unordered collection of its words. Order is discarded; what survives is which published points a route uses, and that is already enough to let the model exploit route structure rather than aggregate distance alone. For instance, two routes on the same city pair may differ by under one per cent in distance and in planned fuel, and so be indistinguishable on every indicator, while only one of them crosses a volume whose identifier the model has met in thousands of earlier revisions; the token set carries that fact and the aggregate distance cannot. Stay pairs enter the fitting set with a group weight w, and the weight is swept over the values 10, 30 and 100; the selection of $w = 10$ is discussed with the change-bias results in Section 6.3.

Once trained, the model has still to be connected to a decision, and a raw score is not sufficient for that. Let us consider a real pair from the held-out quarter, on which the model scores the two routes -2.82 and $+0.90$. Neither number means anything on its own: the score has no units, and the same two numbers on another flight would carry no comparable meaning. What can be read is their difference, here 3.7 in favour of the second route. Even that is not yet a decision, because nobody in an operations room knows what a gap of 3.7 is worth. Calibration is the step that turns the gap into a statement of the form “about eight times in ten, the route the model prefers is the one the airline goes on to file”.

The conversion is a monotone mapping from score gap to observed rate, fitted on the tuning month alone (Zadrozny and Elkan, 2002) and applied unchanged to the evaluation months, and Table 3 shows what it does and what it is worth. A gap of half a point maps to a probability of about 0.72, against which the higher-scoring route in that band was the one the airline filed in 68.5 % of held-out cases; the 3.7 of the example above maps to 0.99, and in that band the model was right 97.4 % of the time. The stated probability runs a little above the observed share through the middle of the range, by at most 4.4 points, and the paragraph below reports that gap as one number. It is the probability, never the gap, that the proposal policy of

Table 3: From score gap to acceptance probability, on the held-out quarter. The first column is the gap between the two routes' scores, the third the probability the calibrator assigns to the higher-scoring route, and the fourth the share of those cases in which that route was in fact the one the airline filed. The two track each other, the stated probability running above the observed share by 4.4 points in the worst of these six bands and by 1.7 points across them, weighting each band by the pairs in it (the text quotes 1.8, the same average taken over fifteen finer bands). The model is mildly over-confident in the middle of its range, and a channel that read its probabilities as exact would slightly overestimate how often it is right.

score gap	pairs	calibrated probability	airline filed that route [%]
below 0.5	144 518	0.551	54.7
0.5 to 1	49 784	0.718	68.5
1 to 2	40 524	0.876	83.2
2 to 3	19 520	0.958	93.3
3 to 5	18 645	0.988	97.4
above 5	8 729	0.997	99.8

Section 7 compares against its threshold, and it is calibration that makes a precision target meaningful at all, since without it one can rank proposals but cannot decide how many of them to make.

Three mappings were compared on the tuning month, namely the raw sigmoid of the score gap, Platt scaling (a two-parameter sigmoid fit) and isotonic regression (a stepwise monotone fit), of which isotonic regression reached the lowest tuning-month log loss and is what the pipeline uses. Miscalibration is common in flexible models fitted this way (Guo et al., 2017), and this model runs the other way on a different criterion from the one that chose the calibrator, in that on the held-out quarter its own raw sigmoid was better calibrated than the isotonic fit selected to correct it: the expected calibration error (the mean gap between stated probability and observed rate) is 0.9 percentage points for the sigmoid against 1.8 for the isotonic fit. The selection was made on the tuning month before the evaluation months were touched and is reported unrevised. The reading the authors draw from it is that on this archive the fitted mapping is a safeguard rather than a correction, and that a deployment could reasonably use the sigmoid instead.

6. Predictive performance

This section reports how well the model predicts, and then decomposes that figure. Accuracy is set first against coverage and against the simple rules a channel could use instead, and then resolved into the strata where it lives, the design choices that earn it, the errors that remain and the features the model turns out to weigh.

6.1. Accuracy, coverage and baselines

On the held-out quarter the model identifies the airline's chosen route in 68.1 % of 281 720 pairs (95 % interval 67.9–68.3, or 67.7–68.6 under city-pair clustering), against the 50 % of random choice. Reassuringly, the tuning month scores 67.5 %, and there is therefore no sign of overfitting to it.

Accuracy alone understates operational value, because the model is informative about its own uncertainty. Fig. 4 reports accuracy when the model is permitted to abstain on those cases in which the two routes score close together. Let us call the share of flights that the model is allowed to answer its coverage, the remaining flights being left alone. On its most confident fifth of the cases the model is right 94.9 % of the time, and a proposal channel is not obliged to answer every flight: it operates at the left of this curve.

The curve is one reference; the other is what simpler methods achieve on the same held-out quarter. Table 4 sets out the whole ladder, from a coin toss to the model of this paper, so that a reader can see where the accuracy is bought and judge whether the complexity earns its place.

Every rule in that table preferring the route that is *cheaper* on a planning indicator scores below chance: less planned fuel identifies the airline's choice in 45.0 % of pairs, less flight time in 46.8 %, shorter distance in 46.2 % and lower en-route charges in 47.2 %. Filing behaviour on revised flights does not merely fail to minimise planning cost, it runs against it more often than not, for the structural reason that Table 5 sets out below: most revisions buy something, usually relief from delay, and pay for it in fuel and time. A proposal system built on "offer the cheaper route" and nothing else would therefore disagree with the airline's own subsequent behaviour on most revised flights, which is exactly the finding that Section 7 turns into a design constraint.

A consequence for the baseline follows immediately, since a binary rule that scores 45.0 % is a rule that scores 55.0 % when its sign is reversed: the strongest single-feature baseline available on this archive is "prefer the

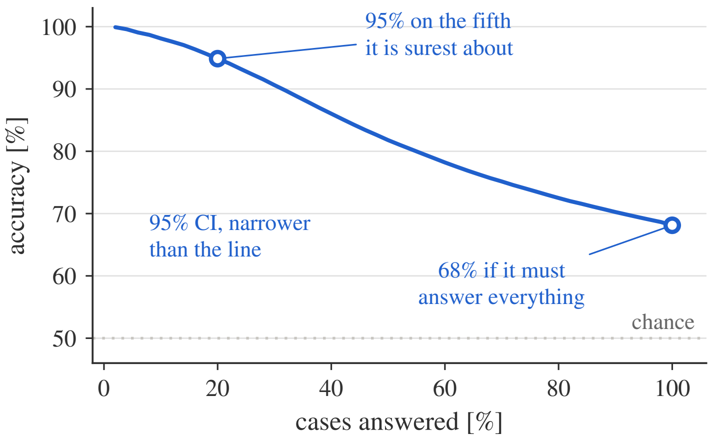

Figure 4: Accuracy against coverage on the held-out quarter. The horizontal axis is the share of cases the model is required to answer, lowest confidence dropped first; the vertical axis is the share of answered cases it gets right. Moving left, the model answers fewer cases and is right more often: 95 % of the fifth it is surest about, against 68 % when it must answer everything. A "case" here is one pair of routes, and answering it means naming one of the two. The dotted line is chance; the 95 % confidence interval is drawn but is narrower than the line itself at every point. It is this curve, rather than any single figure on it, that characterises the system.

route that burns *more* planned fuel", at 55.0 %, and not the delay rule at 53.2 %. The model's margin over the best rule built on any one indicator is accordingly thirteen points rather than fifteen. That inversion is not a rule anyone would deploy, since it tells an airline to burn more fuel for no stated reason, but it is the right bar, and it is the bar the model clears. The delay rule remains the instructive one: it is right 71.6 % of the time, but only ever decides on the 14.6 % of pairs where delay differs at all, and scoring the rest as coin tosses gives $0.146 \times 71.6 + 0.854 \times 50 = 53.2\,\%$. Delay avoidance is real, the model must and does learn it, and it accounts for only a small share of the archive.

The learned baselines settle the remaining question, namely whether the additional complexity is doing any work. A logistic model given the four cost indicators as within-pair differences reaches 54.9 %, and the attributed delay and the count of regulations take it to 56.4 %, where eight further numeric features leave it unchanged. Adding the categorical context takes it to 57.5 %, at which point it has seen every input the model of this paper sees except the waypoint sequence. The gradient-boosted model reaches 68.1 %

Table 4: Every baseline on the held-out quarter, 281 720 pairs. A single-feature rule decides only where both routes carry a value for its indicator and the two values differ, and is scored as a coin toss on the rest; the "decides" column gives that share. A route under no regulation carries no delay figure rather than a zero, which is why the delay rule decides on fewer pairs than the share on which the revision changed the delay in Table 5. The logistic models are fitted on the same 1 134 000 training pairs as the model of this paper, with the same within-pair encoding, and their regularisation strength is selected on the tuning month.

baseline	inputs	decides [%]	accuracy [%]
coin toss	0	–	50.0
prefer less planned fuel	1	97.9	45.0
prefer shorter distance	1	81.1	46.2
prefer less flight time	1	96.7	46.8
prefer lower route charges	1	53.6	47.2
prefer *more* planned fuel	1	97.9	55.0
prefer less attributed delay	1	14.6	53.2
logistic, cost indicators	4	–	54.9
logistic, and delay and count	6	–	56.4
logistic, all numeric	14	–	56.4
logistic, and categorical context	22	–	57.5
this paper's model	23	–	68.1

on those same inputs plus the route text, so roughly eleven points come from the nonlinearity and the route representation together rather than from the choice of features.

That margin establishes that route choice carries structure that neither a cost rule nor a linear index in the same inputs recovers, which was by no means self-evident, but it does not establish that 68.1 % is anywhere near the ceiling. The residual splits in two, and the first part is not information at all, because the two routes of a pair are very often nearly the same route. Where they share more than nine tenths of their waypoints the model scores 61.9 %, against 74.9 % where they share less than a quarter, and on a pair whose two routes differ by a single kilogramme of planned fuel there may be no reason for the airline to prefer either. The rest is a property of the decision rather than of the archive, since a route is chosen by a person, and by different people on different days, from information that no route-level

Table 5: What each revision did to the flight's attributed ATFM delay, on the held-out quarter, and how well the model reads it. Fuel and time differences are the new route minus the one it replaced, so a positive number is a cost the airline accepted. Intervals are Wilson 95 % intervals. This table is keyed on the delay itself, whereas Table 2 is keyed on whether a regulation was attributed to a route. A regulation can be attributed and leave the delay at zero, which is why 71.0 % of this quarter's pairs touch no delay while only 63.3 % of the same pairs carry no regulation (Table 6).

what the revision did	pairs	share [%]	median Δfuel [kg]	new saves fuel [%]	accuracy [%]
no delay either route	199 932	71.0	0	47.7	64.0 [63.8, 64.2]
removed the delay	35 890	12.7	103	26.8	88.2 [87.8, 88.5]
reduced the delay	21 202	7.5	56	31.8	81.5 [81.0, 82.1]
left the delay alone	10 234	3.6	1	45.5	60.4 [59.5, 61.4]
increased the delay	9 189	3.3	−2	52.2	61.0 [60.0, 62.0]
took delay on	5 273	1.9	−10	53.2	60.9 [59.5, 62.2]

record holds: how much duty time the crew has left, a swap of departure slots between the airline's own flights, a company routing policy, a judgement about weather that has not yet materialised. Two competent flight planners facing the same situation may reasonably file differently, and neither of them is making a mistake. Perfect prediction would require that route choice be a deterministic function of what the network records, and there is no reason to expect that it is.

6.2. Accuracy by stratum

The 68.1 % headline is an average over decisions of very different kinds, and the first question a reader should ask is what kinds of decision the archive actually contains. Table 5 answers that and the performance question together, on one population: it sorts the held-out quarter by what the revision did to the flight's attributed ATFM delay, and reports how often the model called each sort correctly.

Seven revisions in ten touch no delay on either route, and the median such revision moves the flight to a route costing the same planned fuel to the kilogramme: this is ordinary flight-plan housekeeping, and the model reads it at 64.0 %, which is well above chance and far below what it reaches on the delay strata. The two rows in which the airline bought relief from delay are the ones the model reads well. When a revision removes the attributed delay outright, one revision in eight, the model identifies the chosen route 88.2 %

of the time, and it is worth noticing what that relief cost: the median such move burns 103 kg more fuel, and only 26.8 % of these moves are cheaper on fuel at all. Reducing rather than removing the delay costs a median 56 kg and is read at 81.5 %. The three remaining rows, in which the delay stayed put, grew, or was newly taken on, together account for 8.8 % of the quarter and are read at about 61 %.

The shape of that answer is what an operational reader most needs, and it divides cleanly in two. The model is good, at 81 to 88 %, precisely where an airline is trading fuel against delay and the stake is large, and weak, at 60 to 64 %, precisely where the two routes cost almost the same and the choice between them carries little consequence either way. That is a defensible division of labour in general rather than a defect, since it is the first kind of decision that a proposal channel is built to reason about. It is also, however, exactly the wrong way round for the channel of Section 7, which proposes on fuel and therefore speaks precisely where the model is weakest, and Section 7 takes that tension up.

Accuracy also climbs steeply with the size of the delay on the abandoned route: 63.9 % where that route carried none, 70.3 % up to a quarter of an hour, 79.3 % up to half an hour and 84.5 % from half an hour to an hour, where the climb stops: beyond an hour it is 83.5 % (Table 6). The pattern is exactly what an economic reading predicts: the larger the delay, the clearer the incentive, and the easier the decision is to predict.

The final block of Table 6 reads strangely at first and repays attention. The model is right 79.9 % of the time on the pairs where none of the three costs an airline pays for directly is cheaper on the newly filed route, and only 53.9 % on those where all three are. That first row, the 90 961 pairs on which no indicator prefers the new route, is not a population that moved to a strictly worse route, because it also holds every pair on which the two routes are simply identical on an indicator, and route charges are identical on 46.4 % of the quarter. Restricting it to the 38 990 pairs that are strictly worse on all three, which is 42.9 % of the row, sharpens the contrast rather than softening it, because the model is right on 86.3 % of them.

The 79.9 % and the 53.9 % are one result, and the record explains it only in part. A move the indicators all argue against is more often a move made for a reason they do not carry: a regulation is attributed to one of the two routes on 42.2 % of the 90 961 pairs, and on 63.5 % of the strictly worse 38 990, against 33.7 % of the 24 736 pairs the indicators all favour. Regulation exposure therefore differs between the first row and the last by

Table 6: Accuracy by stratum on the held-out quarter (Wilson 95 % intervals, the standard interval for a proportion). The first block splits on whether a regulation was attributed to each route, which is the convention of Table 2 and not the delay-keyed one of Table 5, so the counts differ from that table's. The third block counts how many of the three cost indicators favour the newly filed route; the count is over the quantities an airline pays for directly, so distance, which is a driver of those costs rather than a bill in its own right, is not part of it. An indicator counts as favouring the new route only where that route is strictly cheaper on it, so a pair on which the two routes are identical on an indicator, which happens on 46.4 % of the quarter for route charges, falls towards the first row rather than the last. The block is keyed on agreement with the route the airline actually filed, which means that a rule simply following the indicators is wrong by construction in the first row and right by construction in the last; the rows measure what the model adds to the indicators rather than how the two compare.

stratum	pairs	accuracy [%]
neither route regulated	178 370	64.0 [63.8, 64.2]
old route only regulated	30 953	88.0 [87.6, 88.3]
both routes regulated	67 587	70.6 [70.2, 70.9]
new route only regulated	4 810	58.6 [57.2, 60.0]
delay on old route: none	205 205	63.9 [63.7, 64.1]
up to 15 min	19 672	70.3 [69.7, 71.0]
15 to 30 min	18 283	79.3 [78.7, 79.9]
30 to 60 min	21 231	84.5 [84.0, 85.0]
above 60 min	17 329	83.5 [82.9, 84.0]
cost indicators favouring new: 0 of 3	90 961	79.9 [79.7, 80.2]
1 of 3	87 471	70.0 [69.7, 70.3]
2 of 3	78 552	56.8 [56.5, 57.2]
3 of 3	24 736	53.9 [53.3, 54.5]

8.5 points, and nothing in the archive connects that difference to the 26 points of accuracy which separate them, so the size of the contrast is observed rather than explained. Its direction is not in doubt. The model contributes most precisely where cost logic fails, and least where cost logic was already right, which is the correct division of labour for a system intended to complement cost models rather than replace them.

One consequence of the regulation strata, the first block of the same table, is worth stating before the headline is read anywhere else. The held-out quarter is not a random slice of the archive: 63.3 % of it carries no regulation on either route against 70.6 % of the eighteen months, and that unregulated

Table 7: Screening ablations. Each variant is retrained from scratch at a deliberately reduced setting, depth 6 and 800 boosting rounds, which reaches 60.6 % on revision pairs; the setting is cheap enough to run every variant and is not the configuration reported elsewhere in this paper. Values are the change in revision accuracy against that reduced baseline of 60.6 %, and against nothing else; they are not headline accuracies and are not differences from the 68.1 % model. A positive value means the variant scores *higher* on revisions than the adopted feature and weight configuration does at that same reduced setting, which for the stay-pair correction is the price paid for the change-bias reduction reported below rather than an argument against it. At full scale that removal is worth 0.7 points on the sweep's own runs (Table 8) and 0.8 against the adopted model, not 3.7. Differences are significant under McNemar's test, which asks whether two models' disagreements on the same flights fall asymmetrically rather than by chance, at $p < 0.001$ except the waypoint text at screening scale ($p = 0.58$). The monotone constraint (the score forced never to rise as the number of regulations on a route grows) costs a fifth of a point, which is why none is imposed. A vertical-only revision changes the requested cruise level and leaves the horizontal route untouched, so its two routes are identical and the pair is unanswerable; such pairs are excluded from the archive, and the row shows what including them would cost, which unlike the other four rows is a change of test set rather than a change of model.

variant	Δ accuracy [pp]
without within-pair KPI normalisation	−10.0
without stay-pair correction	+3.7
with monotone constraint	−0.2
without waypoint text	+0.03
including vertical-only revisions	−2.0

stratum is the hardest of the four. Reweighting the four regulation strata to the archive's own mix gives 67.1 % against the 68.1 % headline, so about one point of the headline is the composition of the quarter rather than the skill of the model.

6.3. Ablation and change bias

Table 7 reports screening ablations, each configuration removed in turn and the model retrained, and one of the five dominates the rest.

Expressing the indicators relative to the other route in the pair is decisively the most important choice, since removing it costs nearly ten percentage points: it is what converts four absolute magnitudes into a comparison. The waypoint-text result requires care, and illustrates why screening ablations can mislead. Interestingly, at screening scale (the reduced setting of Table 7) the text feature appears worthless, and yet retrained at full scale it is worth +1.4 percentage points (68.2 against 66.8, the 68.2 being a separate run of the adopted configuration, whose seeds span 0.17 points of accuracy), the reason being that a bag-of-words representation spread over thousands of rare tokens cannot repay its cost in 800 shallow rounds. It is a real but secondary contributor, worth about a tenth of the model's margin over the best single rule of Section 6.1. Yet attribution ranks the same feature first, an apparent tension that Section 6.5 resolves.

A bag of tokens is a deliberately crude way to carry a route, and a more principled alternative was tried before it was settled on. Each route was recast as a sequence of segment descriptors invariant to translation, rotation and scale, each leg encoded by its change of bearing relative to the great circle between origin and destination together with its length as a fraction of the whole, and a neural encoder was trained on that representation under the same pairwise objective. On its own it did worse than the tokens, and supplied alongside them it was worth a fraction of a point, far too little to repay a second model and a waypoint-coordinate service in the serving path. A linear classifier fitted on the encoder's own representation separated the preferred route from the abandoned one barely better than chance, which points at the geometric input rather than the training objective as the limitation. The likely reason is that what the token representation carries is not geometry: an identifier records which published structure a route uses, and two geometrically similar routes through different structures are not interchangeable to an airline, so the simpler representation was kept. One qualification attaches to that comparison and to no other number in this paper. Those trials predate a repair to the archive, which had keyed pair completeness on an identifier that is reused from one day to the next and had therefore dropped about a quarter of the valid pairs; every accuracy reported here is computed after that repair and the trials are not, so only their ordering is claimed.

The change bias is measured on the held-out stay pairs, and one thing has to be settled before the figure can be read. Section 4.2 established that regulation status identifies the answer on every stay pair, so a model could in principle score well on them by reading the regulation and nothing else. The question is whether this one does, and the archive answers it directly, because it contains a second population with the same regulation signature and the opposite label.

That second population is the escapes, in which the abandoned route is the regulated one, so that the right answer is to prefer the unregulated route; in a stay pair the kept route is the regulated one, so the right answer is to prefer the regulated route. A rule that reads regulation status alone must therefore score 100 % on one population and zero on the other. The model scores 88.0 % on the 30 953 escapes of the held-out quarter and 90.0 % on its 1 093 stay pairs, and it does so at a cost of 0.8 points of revision accuracy, 68.1 % against the 68.9 % of a model trained without stay pairs. No reading of regulation status produces that pair of numbers, since telling the two situations apart requires the route and the flight around it, which is precisely the understanding the stay pairs were added to instil, so the stay figure can be read as it stands.

A model trained on revisions alone prefers the alternative route in 65.2 % of the cases where the flight demonstrably kept its own, and so is right on the remaining 34.8 %, because revisions teach it that a regulated route is one to escape and it carries that lesson into a population where the opposite holds. Adding stay pairs at group weight 10 turns 34.8 % into 90.0 %, at the 0.8-point cost on revisions noted above (the 3.7 points of Table 7 are measured at the reduced screening setting, not at full scale).

Two bounds on that result remain, and neither is repaired by how it is measured. The first is the size of the stay set, 1 093 pairs in the test months against 281 720 revisions, which puts the interval around 90.0 % at $[88.1, 91.7]$. The two routes of a stay pair also belong to two different flights, which no revision pair does, so a model could learn to recognise the construction itself rather than the preference it encodes. The balanced stay set of Section 9 would settle both bounds properly.

The weight itself was chosen by the sweep of Table 8, which trains one model per value. Raising it from 10 to 100 buys 7.3 further points of stay accuracy and costs 1.5 points on revisions, which is a poor trade, and $w = 10$ is used here because it takes nearly all of the available correction at the smallest cost. That sweep was read on the held-out quarter, which is not the discipline Section 4.4 states for every other choice in this paper, so it was re-run with the tuning month scored alongside: the tuning month selects the same weight, at 67.5 % there against 67.1 and 66.6 for 30 and 100. One caveat on reading these figures: the sweep's own $w = 10$ run is a separate training run from the adopted model and lands close to it, 90.3 against 90.0 on stay pairs and 68.2 against the 68.1 headline. Four models of the adopted configuration differing only in their random seed span 0.17 points of revision accuracy on the same quarter, which is the run-to-run variation a difference of that size has to clear.

Table 8: The stay-weight sweep, one model trained per value. The stay column is accuracy on the held-out stay pairs and the last column is the price paid on revisions. With no stay pairs at all the model scores 34.8 % on them, far below the 50 % of a coin toss, because it has learned from revisions alone that a regulated route is one to leave and stay pairs are exactly the population where that is wrong. The trade turns bad above a weight of 10: 7.3 further points of stay accuracy cost 1.5 points of revision accuracy. These are the sweep's own runs; the adopted model of this paper scores 90.0 and 68.1, and four seeds of that model span 0.17 points of revision accuracy.

stay weight	stay [%]	revisions [%]
none	34.8	68.9
10	90.3	68.2
30	95.2	67.7
100	97.6	66.7

6.4. Error analysis

Fortunately, errors concentrate where the model is already unsure. Only 5.6 % of them fall among the quarter of pairs whose two routes score furthest apart, against the 25 % that would fall there if error were independent of confidence. The same pattern appears from the other end, in that 27.7 % of the errors are near-ties, in the sense that the two scores land within one fourth of the archive's median gap. Errors therefore concentrate at low confidence, which is the property that makes confidence filtering work.

The confident errors are nevertheless instructive, and one example from the held-out quarter is worth following. A business jet on a long-haul flight re-filed off a route carrying 124 minutes of attributed delay onto a route that carried no delay at all, saving 152 kg of fuel and about a minute of flight time; every cost indicator the model sees favoured the move, and so did the delay, and yet the model confidently scored the abandoned route higher. The flight-level context is identical within a pair by construction, and here every route-level quantity except the waypoint sequence favoured the move, so the preference came from that sequence, presumably a learned structural preference that overrode the rest. Confident errors are therefore not all of the hidden-information kind, meaning decisions driven by facts that appear in no route-level record, such as a slot swap (an exchange of departure slots between an airline's own flights) or a curfew (a night-time closure at an airport).

Opportunely, such cases cost less than their share of the error rate would suggest, because a proposal channel would have said nothing at all here: the alternative that it would have offered was the one already filed. The expensive errors are the opposite ones, in which the model confidently proposes a route that the airline declines, and the change-bias measurement reported above is the closest available proxy for their rate.

6.5. Feature attribution

A ranking model that cannot be interrogated is of little use to an operational user who is asked to trust it. Shapley attributions were therefore computed over the held-out quarter: each feature is credited with the share of the score that is attributable to it, averaged over every order in which the features could have been supplied to the model (Lundberg and Lee, 2017). Because the model decides on a comparison rather than on one route, what is reported here is the *difference* between a feature's attribution on the two routes of a pair, averaged in magnitude over the quarter (Fig. 5). That is a ranking not of what correlates with acceptance, but of what moves this particular model's decisions.

The waypoint sequence is the largest single term in that ranking, at 0.40 against 0.26 for attributed delay and 0.20 for planned fuel, so route structure carries information about the choice that no individual cost indicator carries. The claim should be made at exactly that width and no wider: the four cost indicators taken together sum to 0.52 and therefore outweigh the route text (the figure's bars are rounded to two decimals and sum to 0.51). What the attribution supports is that route structure is information a cost model does not have, not that it dominates the decision.

Attributed ATFM delay comes next, moving the score more than planned fuel does. Why it does is not something these figures settle: the ranking is of what moves this model, and a tree ensemble responds to how a feature is distributed rather than to the units it is measured in. What can be said is that nothing here shows airlines find delay the easier of the two to price. If anything the reverse holds, since fuel has a market price and the cost of delay is particular to the airline, which is exactly why the trade between them has to be learned rather than derived.

One entry in the ranking is a caution rather than a finding. The time remaining to departure appears fourth, at 0.18, even though the encoding of Section 4.3 makes it identical on both routes of every pair. A feature with the same value on both sides cannot by itself separate them, and this one does

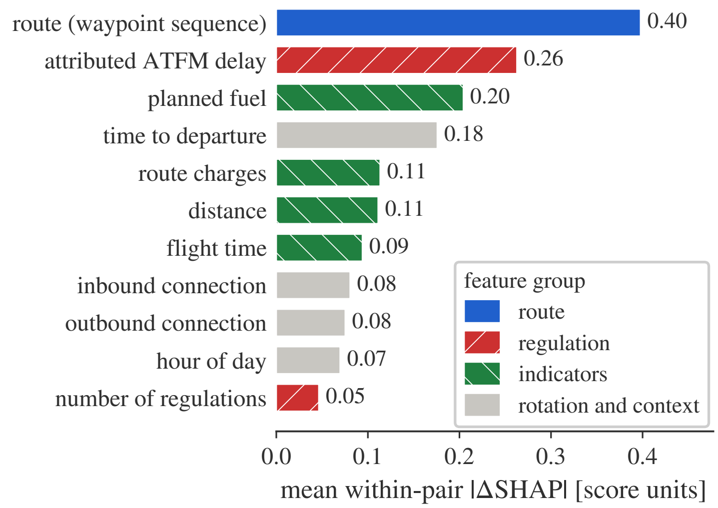

Figure 5: What moves a decision: the mean absolute *difference* between a feature's Shapley value on the two routes of a pair, over the held-out quarter, in units of the ranking score, with each bar's value printed beside it. The eleven features that move the score most are shown, of the twenty-three the model sees; fill colour and hatching group them, with the rotation and context groups of Table 1 drawn together. The waypoint sequence is the largest single term, ahead of every individual cost indicator, though the cost indicators taken together sum to more. A large attribution is not the same as a large ablation cost: removing the waypoint sequence and retraining costs 1.4 points of accuracy, and Table 7's +0.03 is a screening-scale artefact. The two measure different things, as the text below explains.

not. What the attribution measures here is its interaction with the features that do differ, so that the same fuel difference is read differently three hours before departure than twenty minutes before it. That is a real effect rather than a residue of the shortcut the encoding removed, and the two are worth distinguishing because they look alike in the chart.

Magnitude is only half of an attribution, and the other half is direction. Fig. 6 supplies it for the two features that carry the argument, plotting what each contributes to the score gap against the size of the difference it is describing.

The two run in opposite directions, and both make operational sense. Attributed delay is penalised sharply: where the route the airline filed carried an hour and three quarters less delay than the one it dropped, the delay term alone contributes 2.3 points in its favour, and that contribution falls away

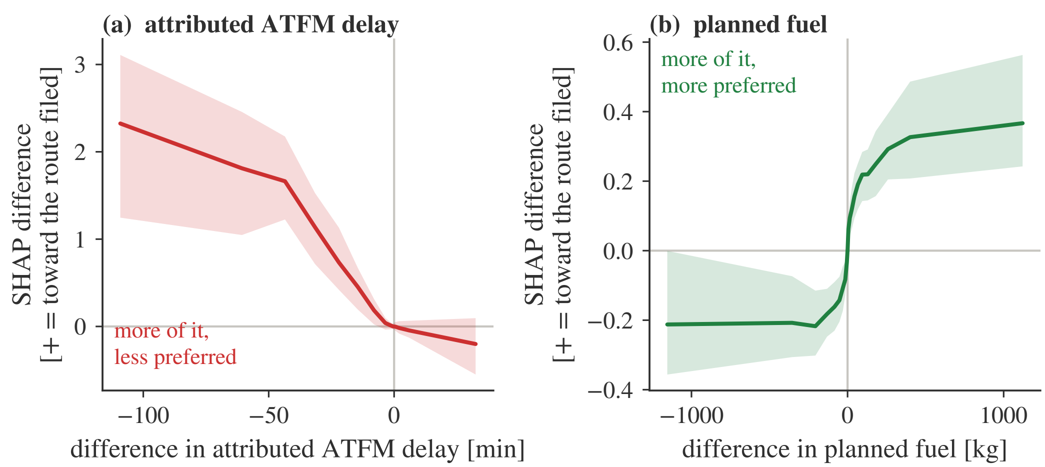

Figure 6: Which way each feature pushes. The horizontal axis is the difference in the feature between the route the airline filed and the one it abandoned; the vertical axis is what that feature contributes to the gap between their scores, as a binned median with the interquartile range shaded. Panel (a): attributed ATFM delay is penalised, strongly. Panel (b): planned fuel is rewarded, weakly, so on this archive the model prefers the costlier route. The waypoint sequence has no scalar value and cannot be drawn this way; time to departure is identical on both routes by construction, so there is no difference to put on the horizontal axis and it is omitted as well. It still carries weight in Fig. 5, where what is measured is its effect on the score rather than its own spread; a feature that is equal on both routes can still change how the model reads the features that differ.

monotonically, reaching slightly below zero when the filed route is the more delayed of the two. Planned fuel runs the other way, and weakly: the fuel term contributes about a fifth of a point *against* a route that burns a tonne less, and about as much in favour of one that burns a little more. The model has therefore learned to reward the more expensive route, which is the same thing the below-chance cost rules of Section 6.1 say about the archive, reached from inside the model rather than from the labels.

The two effects therefore work on very different populations and at very different strengths, in that the fuel term moves the score by a few tenths of a point and does so on almost every pair, whereas the delay term moves it by up to 2.3 points but only on the minority of pairs where a delay exists at all. Their comparable mean magnitudes in Fig. 5 conceal that difference completely. The positive fuel direction, moreover, describes what filing behaviour on *revised* flights looks like rather than what airlines value. A revision usually buys something and pays for it in fuel, so on this archive the costlier route is more often the filed one. A model serving a population that also contains the flights which never revise should not be expected to keep that sign, which is one more reason the change-bias measurement of Section 6.3 matters.

Descending to individual waypoints sharpens the picture, since deleting a single identifier from a route and rescoring shifts the score by up to 2.5 points for the most influential one. Within a route the points that carry that weight are a few of them rather than all of them evenly, which is what a model reading route structure would show and not what a model reading route length would show. Which waypoints those are is not reported, since naming them would identify the airspace and, through it, the operators that use it.

The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy. Attribution measures how much a feature moves the score of the model we have; ablation measures how much worse a model without it would be. Where features are correlated, as route geometry and cost indicators plainly are, the two answer different questions and the second is the one that governs what may safely be dropped.

7. Proposal channel

What a proposal channel is worth turns on how much fuel is at stake and on how confident one can be that the fuel is real. The simulation of this section answers both questions by refusing to count anything that the airline did not subsequently do.

7.1. Policy and operating points

For each pair the channel identifies the route that burns less planned fuel, and proposes it when the calibrated probability that *that* route is the one the airline will file exceeds a threshold τ. The distinction matters, because the cheaper route is often the lower-scoring one. The calibrator maps a score gap to the probability for the higher-scoring route, so when the cheaper route is the lower-scoring one the channel compares one minus that probability against the threshold. That comparison can never succeed at a threshold of a half or above, since one minus a probability of at least a half is at most a half, and the channel therefore proposes only where the cheaper route is also the route the model prefers. Every operating point reported below sits at a half or above, the lowest exactly at it. The threshold τ is selected on the tuning month for a target precision and reported unchanged on the evaluation months, and a proposal is counted as realised when the airline's own subsequent filing chose the proposed route.

A flight is *eligible* under that policy whenever its two routes differ in planned fuel at all, which is 275 907 of the 281 720 held-out pairs, or 97.9 %, only exact ties being excluded. On those eligible flights the cheaper of the two routes is not usually the one the airline went on to file: in 55.1 % of them the fuel saver is the route the airline was in the act of abandoning, so the channel is arguing for staying put at least as often as for moving. A channel that proposed the cheaper route on every eligible flight would therefore be right 44.9 % of the time, which is worse than a coin toss.

That 44.9 % is the figure the model must exceed, and it is what makes the channel a learning problem rather than an arithmetic one. Finding the cheaper route is trivial; the whole task is telling the flights on which the cheaper route is the one the airline wants from the flights on which it is not. Everything Table 9 reports is the size of that gap.

That gap sits awkwardly beside Section 6.2, because the model reads a revision best when the airline was buying relief from delay, and the channel never proposes on those grounds: it proposes on fuel, which is the dimension along which route choice is hardest to call. The channel's answer to that tension is not accuracy but selectivity, since it is free to say nothing, and on four eligible flights in five that is what it does; what carries it from 44.9 % to 79.2 % is the choice of when to speak, not a better reading of the flights it speaks about.

The simulation takes the airline's own next filing as the alternative, and that choice has three consequences. It bounds the evaluation first of all, since the alternative considered for each flight is the other route of its own pair, so the channel is evaluated only on flights that did revise, and Section 9 returns to what that leaves out.

The second consequence is a handicap rather than a flaw, because at the moment a real channel would speak the flight has a route filed and the channel would know which of the two it is. This simulation withholds that: the flight-level attributes are harmonised and the filing time is collapsed, so nothing in the pair marks which route was filed first, and the channel must judge from the routes alone. That is why the 55.1 % of eligible pairs in which the cheaper route is the abandoned one, which is the route on file at the moment a channel would speak, are counted rather than set aside. They are the cases in which a deployed channel should stay quiet, and the model

Table 9: Operating points. The threshold τ is selected on the tuning month for each precision target and evaluated unchanged on the held-out quarter. "Proposed" is the share of the 275 907 eligible pairs on which the channel speaks, the remaining 5 813 of the 281 720 held-out pairs being exact ties on planned fuel, where there is no cheaper route to propose; a channel that spoke on all 275 907 eligible pairs would reach 44.9 % precision. Fuel is the planned fuel saved on the proposals that the airlines' own subsequent filings confirm, per quarter; Section 9 sets out why that is fuel a channel could have surfaced earlier rather than fuel it would create. CO_2 uses 3.16 kg per kg of fuel, applied to the unrounded fuel rather than to the figure printed here.

target [%]	τ	precision [%]	proposals	proposed [%]	fuel [kt]	CO_2 [kt]
60	0.500	65.8	98 783	35.8	24.3	76.9
70	0.525	74.1	74 201	26.9	22.7	71.6
80	0.600	79.2	56 162	20.4	20.1	63.5
90	0.800	88.2	28 756	10.4	13.8	43.6

must determine that it should stay quiet without being told. Withholding the filing order is not a conservative choice that could simply be reversed at serving time, because within this archive that order is very nearly the label itself (Section 4.3). What a deployed channel would know is which route is on file against a candidate that is *not* the airline's own next filing, and nothing here measures what that is worth.

The third consequence governs how every number in this section should be read, because on each flight the simulation counts, the airline found the cheaper route by itself and filed it, some hours after the moment at which the channel would have spoken. The fuel is therefore not fuel a channel would create, but the fuel riding on decisions the model could identify in advance, which measures the scale of what a channel could have surfaced earlier rather than a saving it would add. What a channel would genuinely add lies on the flights that never revised at all, which is precisely the population this archive cannot measure, and Section 9 returns to it.

Table 9 reports the four operating points, and Fig. 7 the continuous trade behind them. Precision is bought with volume, but not one for one: moving the target from 60 to 90 % cuts the share of eligible flights spoken on to less than a third, while giving up 43 % of the confirmed fuel, 13.8 kt against 24.3 kt, because the proposals that survive the higher threshold are disproportionately the valuable ones. At the 80 % point used throughout this paper, the channel speaks on one eligible flight in five and surfaces 56 162

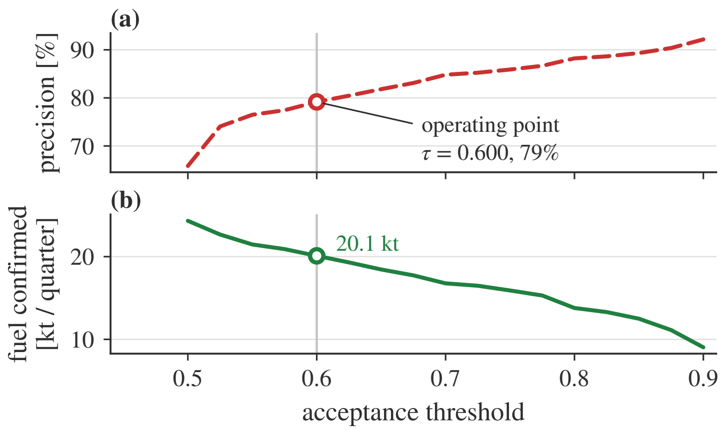

Figure 7: What the channel would deliver, as a function of how selective it is. Panel (a) gives precision, the share of proposals whose route the airline went on to file, and panel (b) the planned fuel saved on those confirmed proposals, both against the same acceptance threshold: raising the threshold buys precision and costs volume, and the confirmed fuel falls with it. The vertical rule through both panels marks the operating point used throughout this paper, which was fixed on the tuning month.

proposals over three months. The airlines' own subsequent filings confirm 79.2 % of them, against the 44.9 % that proposing on every eligible flight would have achieved, and the confirmed proposals carry 20.1 kt of planned fuel, equivalent to 63.5 kt of CO_2, at a mean of 452 kg per confirmed proposal. Annualised, this is 70 to 80 kt of fuel, or 222 to 254 kt of CO_2, the two ends being the two extrapolation bases available: the lower is the per-eligible-pair rate applied to the annual eligible volume estimated from the eighteen-month archive, and the upper is a simple factor of four on the quarter; the corresponding ranges at the other operating points follow the same pattern and reach 85 to 97 kt of fuel at the 60 % target. The median confirmed proposal saves 130 kg and the distribution is strongly skewed: the top tenth of confirmed proposals carries 64 % of the value.

7.2. Concentration of the value

Any first deployment would be a trial rather than a rollout, and a trial small enough to run could still reach most of the value, because the value is concentrated: of the 882 operators that receive any proposal, 6 fly half of the confirmed fuel and 28 fly four fifths of it (Fig. 8). The same holds more weakly by city pair, where 153 of the 10 939 pairs carry half the value, and it does not come from one month: the three test months contribute between

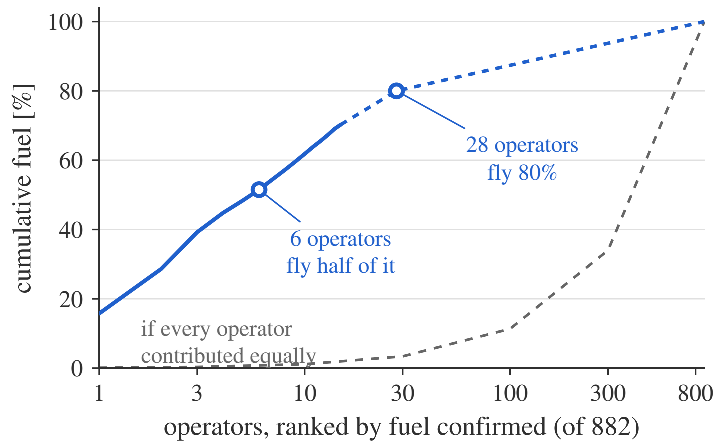

Figure 8: Cumulative share of confirmed fuel, that is, of the planned fuel saved on proposals the airlines' own later filings confirm, against operator rank on a log axis. Six operators fly half of the value and twenty-eight fly four fifths. The grey dashed line is the reference an equal contribution from every operator would trace. The blue curve is measured operator by operator only to rank 15, where it reaches 70.3 % of the value. Beyond that rank the dashed continuation is drawn through two further measured points, 80 % at rank 28 and 100 % at rank 882; those two are counts from the same quarter, and it is the shape of the curve between them that is interpolated.

5.9 and 7.5 kt each. A trial can therefore be small in operators without being small in effect, and the quarter it would be measured against is not the accident of a single month. The counterweight is that the order in which a deployment reaches operators is not neutral. Nothing in the policy withholds a proposal from any operator that qualifies for one, but beginning with six of them decides who benefits first.

7.3. Sensitivity to the bias correction

Every fuel figure above is computed with the stay-pair correction in place, and a fair question is how much of the result it moves, so the whole simulation was repeated with the uncorrected model. At $\tau = 0.600$, the operating point used throughout this paper, the uncorrected variant proposes on 54 139 flights at 81.5 % precision and confirms 20.101 kt, against the corrected model's 56 162 proposals at 79.2 % precision confirming 20.097 kt. At $\tau = 0.500$ the comparison runs the other way on precision, 64.4 % for the uncorrected variant against 65.8 %, and again the two confirm the same fuel, 24.38 against 24.33 kt. The pattern is the same at both points: the confirmed fuel is identical to within fifty tonnes at $\tau = 0.500$ and to four tonnes at $\tau = 0.600$, and the precision differs by 2.3 points at $\tau = 0.600$ and 1.4 at $\tau = 0.500$, in whichever direction the threshold happens to favour. On flights that did revise, in other words, the correction neither helps nor hurts the quantity this section reports. That is the expected result rather than a disappointment, because the correction was never aimed at this population. It is aimed at the deployed one, in which most flights do not revise at all, and the honest summary is that the headline fuel figures are robust to the correction while the argument for deploying the model at all depends on it.

7.4. A worked example

One case from the held-out quarter shows what the channel is for, and is worth following from end to end, because every step is a step that the deployed system would take.

A narrow-body was filed on a medium-haul flight, with no regulation attributed to either route. The system first takes the route then on file and the one alternative offered against it, computes their indicators and subtracts the pair minimum from each, so that what reaches the model is a difference rather than an absolute quantity. What it finds is that the alternative burns 2 016 kg less planned fuel, close to a quarter of the trip fuel, at essentially the same distance and flight time. That margin is chosen because it is unambiguous, and it is far from typical: the median confirmed proposal saves 130 kg, and among pairs whose two routes lie within 50 km and five minutes of each other only about one in a hundred differs by so much. Both routes carry the same flight-level context, namely the same airline, the same aircraft, the same departure hour and the same inbound and outbound connections. Needless to say, nothing that the model sees distinguishes the two routes except where they go and what they cost.

The model then scores each route, and the alternative comes out $+3.7$ ahead. That margin is not yet a decision: it passes through the calibrator fitted on the tuning month, which converts it into an acceptance probability, and that probability is finally compared with the threshold $\tau = 0.600$ that was fixed before the evaluation period began. This case clears the threshold and the policy therefore proposes; had the margin been smaller, the flight would simply have been left alone, since the system is permitted to say nothing, and on roughly four flights in five that is what it does.

The airline's own later filing chose exactly that route. Keep in mind, however, that the airline was told nothing, so this is evidence not that a proposal would have been accepted, but that the model identified, before departure, a route the airline itself went on to prefer. That is the strongest validation available without running the channel, and the form on which every fuel figure in this paper rests.

8. Model ageing

A model that scores well on average but decays from month to month is not deployable, and the main split can only partly address the question: its three test months rule out rapid decay, but they say nothing about the retraining cadence a deployed system would need. This section answers with a second experiment, on the second split of Section 4.4: the same configuration (stay weight 10, within-pair normalisation, waypoint text) is fitted on January to October 2025, tuned and calibrated on November and December 2025, and then evaluated month by month on January through June 2026, so that test month m sits m months beyond the training cutoff. Nothing from 2026 touches the model, the calibrator or the threshold, which is selected on the tuning months for the 80 % precision target exactly as in Section 7.

Training proceeded exactly as for the production configuration, and the threshold selected on the 2025 tuning months for the 80 % precision target was $\tau = 0.600$, the same value the main split produced from its own tuning month three months further on. Table 10 then reports the six months in full, and Fig. 9 the two curves that matter.

Pairwise accuracy starts at 67.2 % one month beyond the cutoff, drifts down to 64.6 % at three months, and then recovers to 68.0 % at six. Interestingly, the worst month is not the furthest one: the curve is U-shaped, and the best of the six months is the last. Two mechanisms plausibly contribute, of which the first is a shift in the mix: the regulated share of the test pairs, meaning those whose newly filed route carries a delay, grows from 5.7 % in January to 22.6 % in June as the summer regulation season arrives, and by June that stratum is the easier one, at 71.4 % against 67.0 % on the unregulated pairs of the same month. The shift does not explain the recovery on its own, however, because the same U-shape appears within the unregulated stratum alone (67.0 % down to 64.7 % and back to 67.0 %). The second mechanism, which we hypothesise rather than measure, is a seasonal re-alignment: June 2026 resembles the June 2025 that sits inside the training window more than the late-winter months resemble anything the window emphasises. Elapsed time and season are confounded by construction in any

Table 10: The year-2025 model evaluated month by month across the first half of 2026 (Wilson 95 % intervals). Coverage here is the channel's, not the abstention coverage of Fig. 4: it is the share of eligible flights on which the channel proposes at the threshold fixed on the 2025 tuning months; the regulated share is the fraction of test pairs whose newly filed route carried attributed delay, which is a third convention, distinct from both Table 2's and Table 5's, and the strata quoted against it in the text are this study's own. It rises through the half year because the summer regulation season arrives, not because the model changes.

month	months past cutoff	accuracy [%]	regulated share [%]	coverage [%]	precision [%]
January	1	67.2 [66.8, 67.5]	5.7	22.2	78.9 [78.3, 79.6]
February	2	66.1 [65.7, 66.5]	5.1	21.3	78.6 [77.9, 79.3]
March	3	64.6 [64.2, 65.0]	5.2	20.4	78.0 [77.3, 78.7]
April	4	65.6 [65.2, 65.9]	8.6	19.8	78.8 [78.2, 79.5]
May	5	66.4 [66.1, 66.7]	16.2	18.7	77.2 [76.6, 77.9]
June	6	68.0 [67.7, 68.3]	22.6	17.3	77.8 [77.2, 78.5]

single such experiment, and the data cannot separate them cleanly; what the experiment does establish is that six months of staleness never cost more than three points. Where the two designs overlap the comparison is direct: a model trained through February 2026 scores 67.5 %, 67.7 % and 69.0 % on April, May and June, so the year-2025 model gives up 1.9, 1.3 and 1.0 points on the same months, and 2.9 points on March against the fresher model's score on its own tuning month. It should be said exactly what that gap measures, because it is smaller than the design might suggest. The two models differ by three months of recency, their data ending in December 2025 and March 2026 respectively, and also by four months of training volume, since the older model is fitted on ten months against fourteen. The one to three points are the joint cost of both, largest at the seasonal antipode, and this experiment cannot separate them. A second measurement is free of that confound, because it compares the aged model only with itself: across the six months of Table 10 its accuracy never falls more than 2.6 points below its own first month, so elapsed time alone, on this evidence, is worth a few points at most.

What happens to the channel over the same half year differs in kind from what happens to the accuracy. At the threshold fixed on the 2025 tuning months, precision on proposals is 78.9 % one month out and 77.8 % six months out, and never leaves the band from 77.2 to 78.9 % across the half year; the

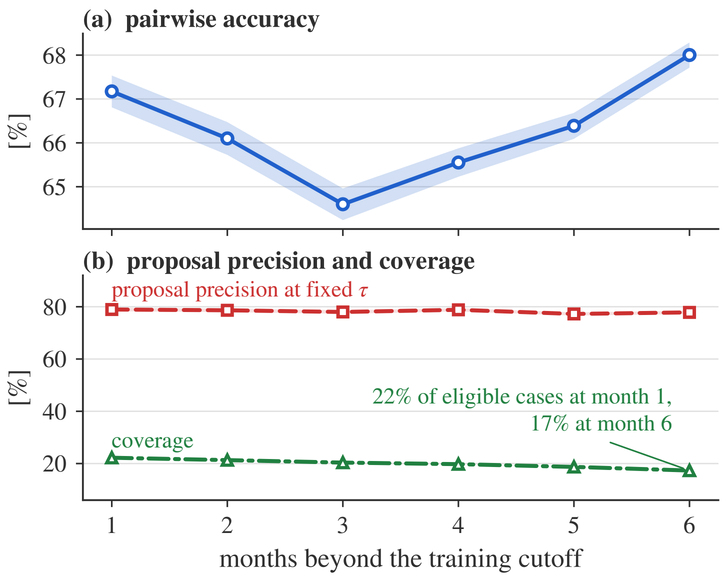

Figure 9: How the model ages, over the six months following its training cutoff. Panel (a): pairwise accuracy dips to 64.6 % at three months and recovers to 68.0 % at six. Panel (b): the precision of the channel's proposals, at the threshold fixed on the 2025 tuning months, never leaves the band from 77.2 to 78.9 %, while the share of eligible flights it proposes on falls monotonically from 22.2 to 17.3 %. Shaded bands are 95 % confidence intervals; coverage is a count over eligible cases rather than an estimate and carries none. The reading is that the ageing model proposes less and errs no more, although part of the coverage fall within this window is seasonal, and the season-matched comparison in the text is the cleaner one.

main split's operating point, tuned on the later month, delivered 79.2 % on its own quarter, and the aged model sits about a point below it on this measure. What moves instead is the share of eligible flights the channel speaks on, which falls from 22.2 % in January to 17.3 % in June. As the traffic drifts away from what the model was trained on, its calibrated confidence clears the threshold less often, and the channel becomes quieter rather than wrong. For a proposal channel this is the correct failure mode, because a wrong proposal costs trust in the way Section 1 describes, whereas a missed one merely leaves fuel where it was. It should be noted that the monthly eligible volume grows by two thirds across the window as summer traffic builds, so part of the coverage movement within the half year is seasonal composition rather than ageing. On the held-out quarter, however, the two designs face the same flights and the season cancels exactly: over April to June the aged model proposes on 18.5 % of the eligible pairs against the fresh model's 20.4 %, at 77.9 against 79.2 % precision. The net cost of those three months of staleness to the channel is therefore 1.9 points of proposal volume and 1.3 of precision, which is the cleanest comparison this data affords, though the two models still differ in training volume as well as in recency.

For a deployment plan the reading is straightforward: retraining quarterly would recover the one to three accuracy points that separate the aged model from the fresh one, of which staleness is one part and the smaller training window the other, and a model left to run for six months beyond its cutoff still held proposal precision inside a band two points wide. The price is quieter operation, and how much quieter depends on which comparison is made: the raw fall across the half year takes the channel from 22.2 to 17.3 % of eligible flights, but the season-matched comparison, in which both models face the same three months, puts the cost of staleness at 1.9 points, from 20.4 down to 18.5 %. Six months and one seasonal transition are the whole of the evidence, so an annual cadence is not something this experiment can recommend, and nothing here reaches a model several years old, still less one serving traffic whose filing behaviour a live channel has itself begun to influence.

9. Threats to validity

The limitations of this study are collected here rather than scattered, in the order of how much they bound the claims.

The label is behaviour, not utility. A revision records what the airline's filing process produced, which folds together its economics, its dispatching software, its habits and its workload. The model learns this compound, and the compound is arguably the right target for a proposal channel, since a proposal is accepted or ignored by the same compound. Two consequences follow. The first is that a revision need not be a choice at all: a route that ceased to be filable has to be replaced whatever the airline would have preferred, and the archive carries no field that separates such a revision from a deliberate one, so a share of the labels that cannot be quantified here records a constraint rather than a preference. The second is that nothing in this paper measures a preference in the economic sense of Samuelson (1938) separately from the process that expresses it, and transfers of the learned function to settings where the process differs (e.g., another region's filing culture) should be expected to degrade.

Flights that never revise. Every revision pair is a flight that changed its plan. A deployed channel would see every flight, and most flights never revise, so the model would be asked a question on which it was evaluated only through the small stay set of Section 4.2. The honest expectation is that precision falls, though not for the reason it is tempting to give. Non-revision has several causes, of which contentment with the filed plan is only one: nothing may have prompted a second look, the dispatch desk may have had no capacity for one, or the flight may have departed before the regulation that would have prompted it was published. The channel's premise is that the second and third of those are common, and this paper measures none of them. The value of the channel on that population is therefore unmeasured in either direction. What the stay pairs do close, as Section 6.3 shows, is the model's tendency to recommend change against traffic that stays, from 34.8 % to 90.0 %. Nor is the model reaching that figure by reading regulation status, since it scores 88.0 % on the population where such a reading would be wrong. Two bounds remain even so, in that the stay set is small, 1 093 test pairs, and its two routes belong to two different flights, so a model could learn to recognise the construction rather than the preference it encodes. That second bound is the one place where this paper compares routes across flights, which is the comparison Section 5 rejects when it argues that a score means nothing except against another route for the same flight. A balanced stay dataset, in which regulation status is independent of the label and both routes belong to the same flight, requires data that is not currently assembled rather than data that does not exist, and is the single most valuable extension of this work. This bound falls harder on the second use case of Section 2.2 than on the first, because an overload release has to rank the flights it finds on the volume, most of which never revise, which is exactly the population on which the acceptance estimates are least reliable. That asymmetry is the reason the second use case is described as enabled rather than demonstrated.

The candidates would come from somewhere else. In this simulation the alternative offered to a flight is the route the airline was about to file. A deployed channel has no such route, and would have to obtain candidates from a generator. That is precisely the distribution Section 2.3 argues is a poor source of *labels*, and the argument transfers uncomfortably to serving time: a model trained on two routes that both passed an airline's own flight-planning system may score a generated route low for reasons that have nothing to do with preference. Nothing in this paper measures that gap, and every figure in Section 7 is conditioned on a candidate the airline had itself produced. A first trial should accordingly draw its candidates from routes that operator has filed before on that city pair, where the concern does not arise, rather than from a generator.

The route is carried as a vocabulary, not a geometry. The waypoint sequence enters the model as a bag of published identifiers, so the 1.4 points it is worth are knowledge of a particular vocabulary rather than of route shape, and two limits follow. Those identifiers are European, so a model fitted here would carry little of that knowledge to another region, which is the same non-transferability that the first of these limitations predicts for the filing process itself. They are also revised from time to time as the route network is republished, so a piece of airspace can reappear under a name the model has never met, with nothing in the encoding to connect the two; such changes are infrequent, and the effect is correspondingly slow, but it is a drift that only retraining repairs. Section 6.3 reports the attempt to replace the vocabulary with a learned representation of route shape, and why it did not succeed. A representation that does not depend on names would nonetheless be what makes the model portable, and it remains the most valuable direction on this axis.

Planned fuel is not burned fuel. Every fuel figure rests on planned fuel, and the two differ by whatever the tactical phase imposes (e.g., a direct routing granted en route, or a cruise level capped below the one requested). Quantifying the difference requires flown trajectories matched to the plans that preceded them, and until that is done the totals of Section 7 should be read as an upper bound on what a proposal channel would deliver.

The counterfactual is absent. On every flight the simulation counts, the airline found the cheaper route by itself, so the 20.1 kt is not fuel that a channel would create but fuel that a channel could have surfaced earlier. What a channel would genuinely add, namely the flights on which the alternative was never considered, is precisely what this data cannot measure. Only a trial in which real proposals are made, and their acceptance logged, can close this gap, and such a trial is also the only way to observe acceptance itself rather than its revealed proxy.

The regulation picture is the binding constraint at serving time. Attributed delay and the regulation picture are among the strongest inputs, and both are properties of a moment rather than of a flight. A score computed against a regulation state that has since changed no longer describes the decision at hand, so the binding constraint at serving time is data currency rather than computation: the scoring step is a single pass through a fixed tree ensemble. Harder, and unmeasured here, is that the alternative route's attributed delay is known in the archive only because the airline went on to file that route. For a route nobody has filed, the delay it would attract is a query rather than an observation, and the error in answering it would propagate into the acceptance probability.

10. Conclusions

Airline route acceptance is learnable from flight-plan revisions at network scale, and the findings of this paper are fivefold. First, a model fitted on 1 134 000 revisions identifies the chosen route in 68.1 % of held-out cases and supports operation at a selected precision rather than at a single accuracy figure: on the most confident fifth of the decisions it is right 94.9 % of the time. Secondly, every rule preferring the route that is cheaper on one of the four planning indicators scores below chance on this archive, so that the strongest simple rule is an inverted one at 55.0 %, and the model's advantage concentrates exactly where cost logic fails: 79.9 % where none of the three costs an airline pays for directly prefers the route the airline moved to, against 53.9 % where all three prefer it. Thirdly, a proposal channel tuned for 80 % precision and delivering 79.2 % addresses 20.1 kt of planned fuel per quarter, validated against what the airlines subsequently filed, four fifths of it flown by 28 of 882 operators, and these figures are robust to the stay-pair bias correction. Fourthly, route structure carries information about the choice that the cost indicators do not: the waypoint sequence is the largest single term in what moves the model's score, though the cost indicators together outweigh it and removing the sequence costs only 1.4 points of accuracy. Fifthly, fitted on ten months of 2025 and watched over the six months that follow its data, the model's proposal precision never leaves the band from 77.2 to 78.9 %, somewhat below the 80 % target it was tuned for, while its accuracy falls at most 2.6 points below its first month before recovering with the season: it ages by proposing less, not by erring more.

A route that burns less planned fuel emits less carbon dioxide, and the reason such routes go unflown is not that nobody can find them but that nobody knows which of them an airline would actually take. That is the gap this paper addresses, and the reason acceptance rather than fuel is the quantity modelled: a proposal an airline declines saves nothing, and a channel that makes many of them is switched off. Ranking candidate routes by likely acceptance is what allows the environmentally cheaper proposals to be aimed at the flights on which they are also cheap for the operator, which is the only version of this idea that both sides have a reason to run.

The number that characterises this system is not 68.1 % but the curve of Fig. 4, because a channel chooses where on that curve to sit, and because ranking a pair correctly and proposing correctly are different quantities of which the second is the operational one: at the threshold used here, the channel proposes on about a fifth of the eligible flights, and 79.2 % of those proposals match what the airline went on to file. Accordingly, any first deployment would sit well to the left of that curve, and would move right only as acceptance is observed.

The second use case of Section 2.2, choosing which flights to ask to reroute when a volume of airspace is overloaded, and in which order, remains enabled rather than demonstrated: it needs the same acceptance probability, and it needs it on a population this archive cannot speak for.

Future work should be directed, in the first place, at assembling a balanced stay dataset in which regulation status is independent of the label, which is what currently bounds the change-bias measurement; in the second place, at matching flown trajectories to the plans that preceded them, so that planned savings can be restated as burned ones; and in the third place, at a shadow-mode trial logging the real responses of airlines, which is the only way to observe acceptance rather than its revealed proxy and which would be the first turn of the feedback loop described in Section 2.2. Last but not least, the horizon experiment of Section 8 gives a deployment plan its retraining cadence, but only a live channel can reveal how preferences shift once proposals themselves begin to influence filing behaviour, and any deployment should therefore expect to re-measure everything this paper measures.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

CRediT authorship contribution statement

Ramon Dalmau: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **David Perez:** Conceptualization, Resources, Validation, Writing – review & editing. **Franck Ballerini:** Validation, Writing – review & editing. **Seddik Belkoura:** Methodology, Validation, Writing – review & editing. **Vincent Taverniers:** Resources, Data curation, Writing – review & editing. **Gratiel Marius Marin:** Resources, Validation, Writing – review & editing. **Robin Deransy:** Conceptualization, Project administration, Writing – review & editing. **Benjamin Cramet:** Supervision, Writing – review & editing. **Grigori Gustin:** Supervision, Project administration, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with drafting and editing the manuscript text and with producing the analysis figures from the study's result tables, and Google's Gemini image model to produce the methodology schematic of Fig. 2. In view of the publisher's policy on the use of generative AI in figures, the authors state that this figure is a schematic illustration of the method whose only numbers are the dataset sizes and settings stated in the text, and contains no data, measurement or result; that it was generated under the authors' direction from a description they wrote; that every label was checked by the authors against the method as described in Section 4 and Section 5, two of them being corrected by hand afterwards, a misspelling and a pair of legend labels set against the wrong tracks; and that the authors defer to the editor on whether a manually drawn equivalent is required. No other figure was created or altered with generative AI: every analysis figure is drawn by the authors' own code from the study's result tables. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study is operational air traffic data held by EUROCONTROL and processed under existing data-handling arrangements; it cannot be made publicly available. Aggregated tables underlying every figure are available from the corresponding author upon reasonable request, in the de-identified form this paper itself uses: operators, city pairs and waypoints appear in them by rank rather than by name, which leaves every number and every ordering unchanged. Tables holding one row per flight are not part of that offer.

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